

Trainings ▾

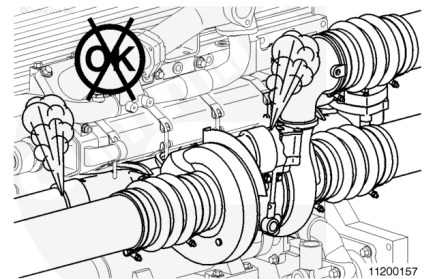
Initial Check

Operate the engine and inspect the exhaust manifold for leaks at the slip joints.

If the manifold is leaking at the slip joints, the slip joint **must** be repaired.

Some ISM engines are equipped with exhaust seal rings located within the slip joints. All ISM CM876 engines are equipped with exhaust seal rings.

Note : For engines not equipped with seal rings, the kit for the front slip joint is different than the kit for the rear slip joint. The outer shells are different, as well as the graphite seal. Only the clamps are similar.



Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠



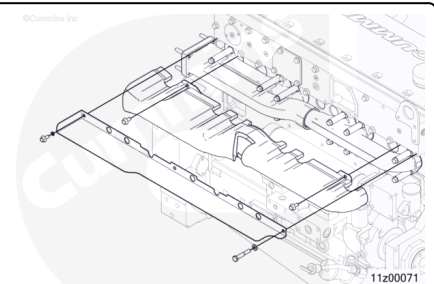
To reduce the possibility of personal injury from hot surfaces, allow the engine to cool before beginning work. Wear appropriate hand and eye protection.

- Remove the turbocharger inlet and exhaust piping. Refer to Procedure 010-033 in Section 10.
(/qs3/pubsys2/xml/en/procedures/35/35-010-033-tr.html)
- Remove the turbocharger. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/35/35-010-033-tr.html)
- Remove heat exchanger. Refer to Procedure 008-053 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-053-tr.html)
- Remove aftercooler assembly (seawater). Refer to Procedure 010-005 in Section 10.
(/qs3/pubsys2/xml/en/procedures/35/35-010-005-tr.html)
- Remove exhaust manifold heat shield. Refer to Procedure 011-032 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-032-tr.html)

Remove

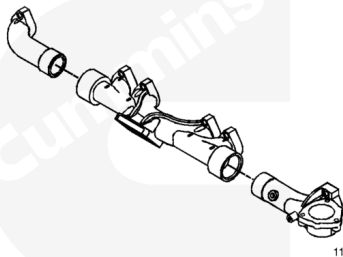
Remove heat shield mounting capscrews from exhaust manifold.

Remove the heat shield.



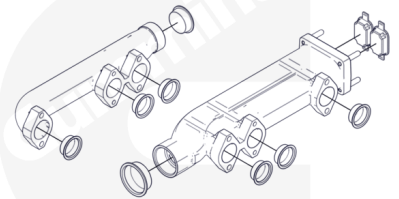
Disassemble

Remove front part of exhaust manifold end section.



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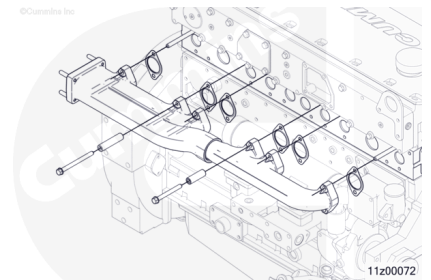
Cover open ports of exhaust manifold with appropriate protective caps or heavy tape.



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Remove two upper capscrews from end sections of exhaust manifold and install two guide studs.

Remove remaining capscrews, exhaust manifold, and discard gaskets.



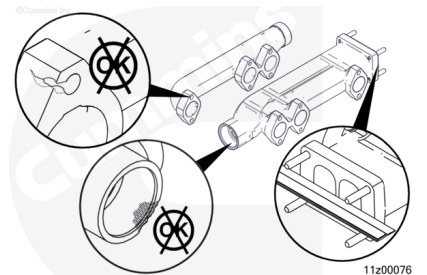
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Clean and Inspect for Reuse

Inspect the parts for cracks or other damage.

Inspect the exhaust manifold slip joint for wear.

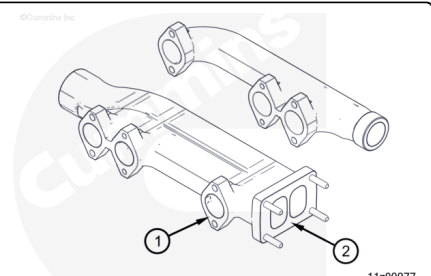
Inspect the exhaust manifold turbocharger mounting surface for warping.



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Measure the mounting surfaces for flatness.

The exhaust manifold mounting surface (1) **must** be flat to within 0.25 mm [0.010 in]. The turbocharger mounting flange (2) **must** have a surface flatness of 0.13 mm [0.005 in].

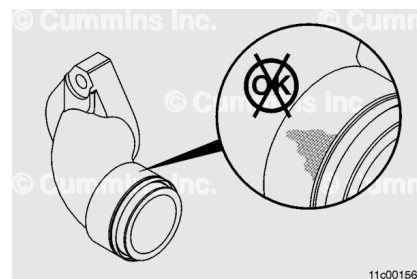


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If these surfaces are **not** within the flatness specifications, see the Alternative Repair Manual, L10 and M11 Series Engines, Bulletin 3810310.

Inspect exhaust manifold slip joints for signs of soot leakage.

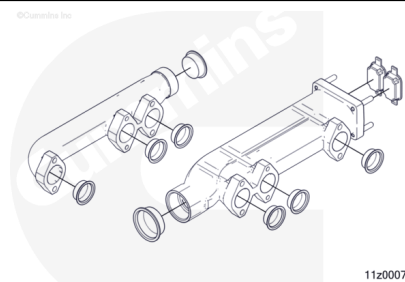
If no evidence of an exhaust leak from the slip joints and there are no visible cracks or damage, do **not** remove the exhaust manifold end sections.



Assemble

Remove any caps or tape covering exhaust manifold ports.

Install exhaust manifold end section to second exhaust manifold section.



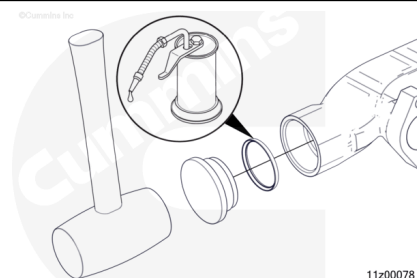
Note : QSM11 engines do **not** have an exhaust manifold seal.

The below steps in the Assemble section do **not** apply to the QSM11 engines

Use a light coat of grease or clean engine oil to lubricate the exhaust seal during assembly onto the exhaust manifold.

Position the seal installer and seal in line with the sealing surface on the exhaust manifold and strike with a lead or a dead blow hammer until the seal is fully seated.

Seal rings, if equipped, **must** remain square to manifold component axes during installation.

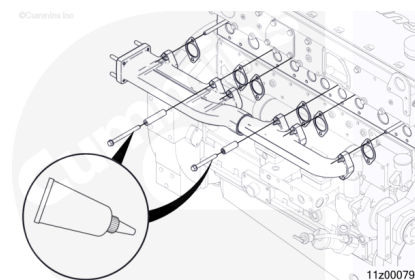


Install

Apply a film of high temperature anti-seize compound to capscrews.

Align the manifold and gaskets using guide studs.

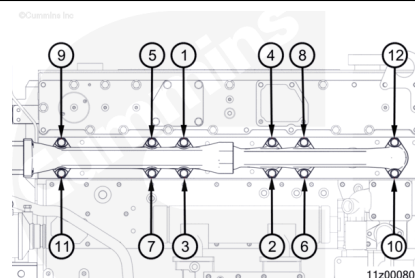
Remove guide studs and hand tighten new capscrews.



Tighten the capscrews in the sequence shown.

Torque Value:

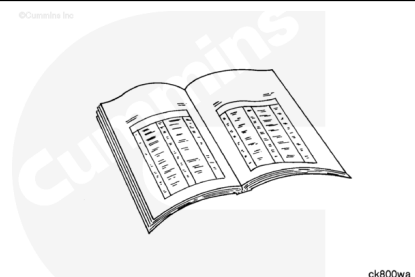
1. 27 N•m [239 in-lb]
2. 47 N•m 35 ft-lb
3. Loosen all capscrews
4. 27 N•m [239 in-lb]
5. 47 N•m 35 ft-lb.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Install the heat shield. Refer to Procedure 008-053 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-053-tr.html>)
- Install the aftercooler assembly (sea water). Refer to Procedure 010-005 in Section 10. (</qs3/pubsys2/xml/en/procedures/35/35-010-005-tr.html>)
- Install the turbocharger. Refer to Procedure 010-033 in Section 10. (</qs3/pubsys2/xml/en/procedures/35/35-010-033-tr.html>)
- Install air piping to turbocharger. See equipment manufacturer service information for correct torque value.
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html>)

- Connect the batteries. See equipment manufacturer service information.
- Operate the engine. Check for air or exhaust leaks.

Last Modified: 03-Jun-2019
